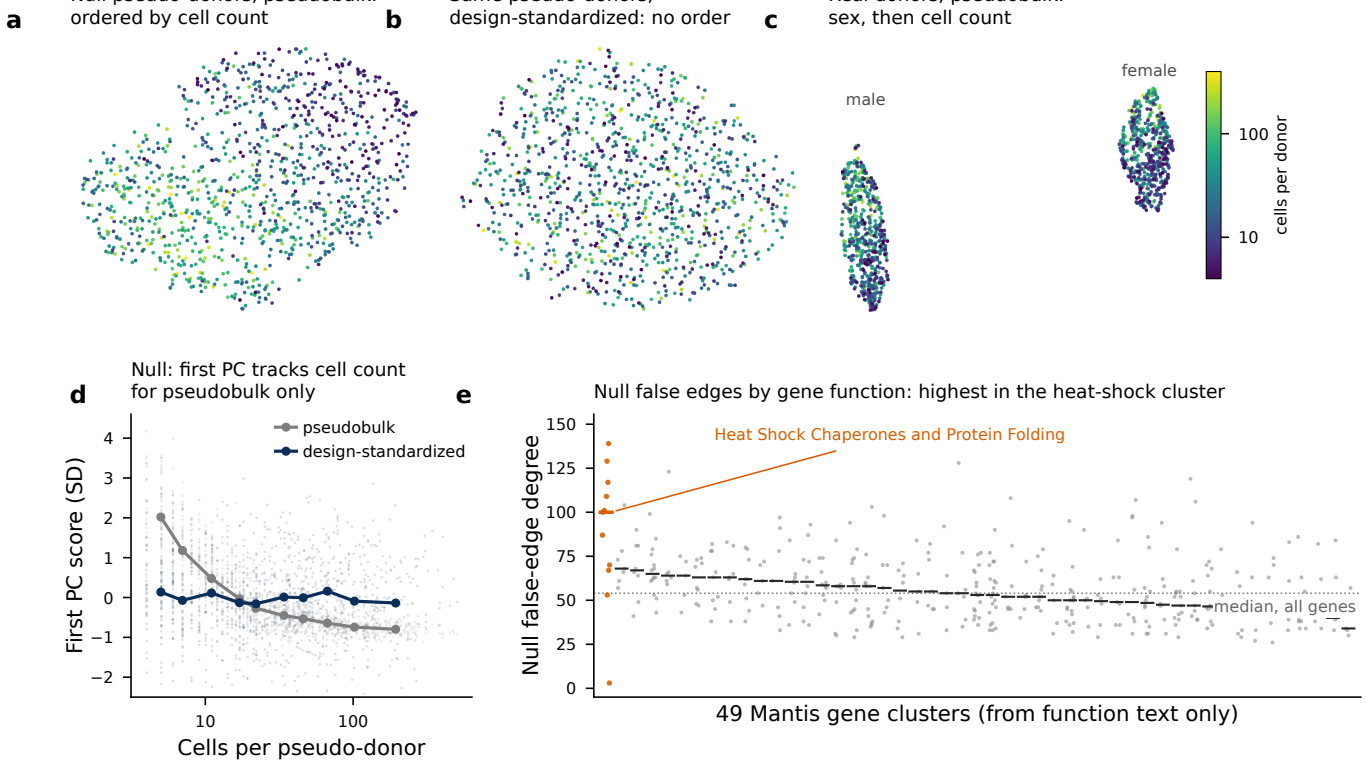

a
Null pseudo-donors, pseudobulk:
ordered by cell count
b
Same pseudo-donors,
design-standardized: no order
c
Real donors, pseudobulk:
sex, then cell count
male
female
100
10
cells per donor
d
Null: first PC tracks cell count
for pseudobulk only
pseudobulk
design-standardized
First PC score (SD)
4
3
2
1
0
−1
−2
10
100
Cells per pseudo-donor
e
Null false edges by gene function: highest in the heat-shock cluster
Heat Shock Chaperones and Protein Folding
Null false-edge degree
150
125
100
75
50
25
0
median, all genes
49 Mantis gene clusters (from function text only)